6.2. Mathematical Expectation

Introduction

An extremely useful concept in problems involving random variables or distributions is that of expectation. Random variables can be characterized and dealt with effectively for practical purposes by consideration of quantities called their expectation. The concept of mathematical expectation arose in connection with games of chance. For example, a gambler might be interested in his average winnings at a game, a businessman in his average profits on a product, and so on. The average value of a random phenomenon is also termed as its Mathematical expectation or expected value. In the following sections, we will define and study the concept of mathematical expectation for both discrete and continuous random variables, which will be used in the following subsection.

6.2.1 Expected value and Variance

Expected value

The expected value is a weighted average of the values of a random variable may assume. The weights are the probabilities.

Definition 6.8

Let X be a discrete random variable with probability mass function (p.m.f.) $p(x)$. Then, its expected value is defined by

$$E(X) = \sum_x x\, p(x) \qquad \ldots (1)$$

If X is a continuous random variable and $f(x)$ is the value of its probability density function at x, the expected value of X is

$$E(X) = \int_{-\infty}^{\infty} x\, f(x)\, dx \qquad \ldots (2)$$

Note

- In (1), $E(X)$ is defined to be the indicated series provided that the series is absolutely convergent; otherwise, we say that the mean does not exist.
- In (1), $E(X)$ is an "average" of the values that the random variable takes on, where each value is weighted by the probability that the random variable is equal to that value. Values that are more probable receive more weight.
- In (2), $E(X)$ is defined to be the indicated integral if the integral exists; otherwise, we say that the mean does not exist.
- In (2), $E(X)$ is an "average" of the values that the random variable takes on, where each value x is multiplied by the approximate probability that X equals the value x, namely $f_X(x)dx$ and then integrated over all values.

DO YOU KNOW?

- $E(X)$ is the center of gravity or centroid of the unit mass that is determined by the density function of X. So the mean of X is measure of where the values of the random variable X are "centered".
- The mean of X, denoted by φ_x or $E(X)$.

Variance

The variance is a weighted average of the squared deviations of a random variable from its mean. The weights are the probabilities. The mean of a random variable X, defined in (1) and (2), was a measure of central location of the density of X. The variance of a random variable X will be a measure of the spread or dispersion of the density of X or simply the variability in the values of a random variable.

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